100G QSFP28 High-TX ZR Data Sheet

Copyright © 2026 FS.COM All Rights Reserved

Applications

- Access and aggregation networks
- Cable TV networks
- Wireless mid-haul & back-haul

Features

- Digital Coherent Optics module, hot-pluggable QSFP28 form factor with extended front section outside the cage (cf. QSFP-DD form factor Type 2)
- IEEE 100G Ethernet (CAUI-4) or ITU-T 100G OTN (OTL4.4) compliant host interface
- 100G optical coherent interface with DP-DQPSK modulation and Staircase FEC per IEEE Std. 802.3-2022 100GBASE-ZR or ITU-T G.709.2
- High Tx output power > 0dBm for compatibility with ROADM line systems
- Full C-band tunable, 50GHz or 100GHz grid with optional Flextune™ automatic wavelength tuning
- Case temperature range 0°C to 70°C (C-temp)
- Typical power dissipation < 6.0W
- Remote digital diagnostics monitoring

Description

The 100G ZR QSFP28 High-Power DCO transceiver supports 100G transmission over distances up to 120km (dispersion limited, optionally extendable to 300km) for edge network applications. On the host side, the module can accommodate IEEE 100GE Ethernet or ITU-T OTN OTU4 signals. The line side coherent interface specifications are aligned with IEEE Std. 802.3-2022 100GBASE-ZR.

Product Specifications

I. Environmental Specifications

Parameter	Conditions		Symbol	Min	Typ	Max	Unit	Notes
Storage temperature			T_S	-40		85	°C	
Case operating temperature	Central office appl. (C-temp)	Long term	T_{OP}	0		70	°C	
		Short term < 96h		-5		75		
Relative humidity	Non-condensing		RH	5		85	%	

II. Interface Modes

- **Host Interface Modes**

Host Interface ID [18]	Host Interface Description [18]	Modulation	Forward Error Correction Code	Nominal Symbol Rate (GBd)	Supported Line Interface IDs [18]
QSFP-ZRH-100G and QSFP-ZRH-100G-S					
65 [9]	CAUI-4 C2M without FEC	NRZ	None	25.78125	68, 192, 193
66 [9]	CAUI-4 C2M with RS(528,514) FEC	NRZ	RS(528,514)	25.78125	68, 192, 193

Copyright © 2026 FS.COM All Rights Reserved

QSFP-ZRH-100G-S only					
57 [10]	OTL4.4 (ITU-T G.709/ Y.1331 G.Sup58)	NRZ	RS(255,239)	27.9525	192, 193

- **Line Interface Modes**

Line Interface ID [18]	Line Interface Description [18]	Modulation	Forward Error Correction Code	Nominal Symbol Rate (GBd)	Spectral Shaping
QSFP-ZRH-100G and QSFP-ZRH-100G-S					
68 [9]	100GBASE-ZR (Clause 154)	DP-DQPSK	Staircase (SC)	27.9525	None
QSFP-ZRH-100G-S only					
192 [11]	OTU4 Long Reach	DP-DQPSK	Staircase (SC)	27.9525	None
193 [10]	OTU4 Short Reach	DP-DQPSK	RS(255,239)	27.9525	None

III. Electrical Characteristics

- **Power & Low Speed I/O**

Parameter	Conditions	Symbol	Min	Typ	Max	Unit	Notes
Power supply - General							
Power supply voltages	Including ripple, droop and noise below 100kHz		3.135	3.300	3.465	V	
Host RMS noise output	10Hz - 10MHz				25	mV	
Module RMS noise output	10Hz - 10MHz				15	mV	
Module supply noise tolerance	10Hz - 10MHz, peak-to-peak	$PSNR_{mod}$			66	mV	
Module inrush	Instantaneous peak duration	T_{ip}			50	µs	
	Initialization time	T_{init}			500	ms	
Power supply - Low power mode							
Power dissipation		P_{lp}			1.5	W	
Power supply current	Instantaneous peak current	$I_{CC,ip,lp}$			600	mA	
	Sustained peak current	$I_{CC,sp,lp}$			495		
	Steady state current	$I_{CC,lp}$			478		1
Power supply - High power mode (with 2.4 ns/nm CD)							
Power dissipation		P_{hp}		5.5	6.2	W	

Copyright © 2026 FS.COM All Rights Reserved

Power supply current	Instantaneous peak current	$I_{CC,ip,hp}$			2480	mA	
	Sustained peak current	$I_{CC,sp,hp}$			2046		
	Steady state current	$I_{CC,hp}$			1978		1
Power supply - High power mode (with 6.0 ns/nm CD)							
Power dissipation		P_{hp}		5.7	6.5	W	
Power supply current	Instantaneous peak current	$I_{CC,ip,hp}$			2600	mA	
	Sustained peak current	$I_{CC,sp,hp}$			2145		
	Steady state current	$I_{CC,hp}$			2073		1
Low speed I/O							
Clock frequency, SCL	Default	f_{SCL}	400			kHz	
	Fast mode+		1000				
Output voltage, SCL and SDA	Output low	V_{OL}	0.0		0.4	V	
	Output high	V_{OH}	V_{CC}-0.5		V_{CC}+0.3		
Input voltage, SCL and SDA	Input low	V_{IL}	-0.3		0.3×V_{CC}	V	
	Input high	V_{IH}	0.7×V_{CC}		V_{CC}+0.5		
Capacitance for SCL and SDA I/O signal		C_i			14	pF	
Total bus capacitive load for SCL and SDA	400kHz clock rate	3.0kΩ pull-up resistor, max.	Cb		100	pF	2
		1.6kΩ pull-up resistor, max.			200		
Input voltage / current, LPMode/TxDis, ResetL and ModSelL	Input voltage, low	V_{IL}	-0.3		0.8	V	
	Input voltage, high	V_{IH}	2.0		V_{CC}+0.3		
	Input current, 0V < V_{in} < V_{CC}	I_{in}	-365		125	μA	
Output voltage, ModPrsL and IntL/RxLOSL	Output low, I_{OL}= 2mA	V_{OL}	0.0		0.4	V	
	Output high, 10kΩ pull-up resistor to host V_{CC}	V_{OH}	V_{CC}-0.5		V_{CC}+0.3		

ⓘ **Notes:** The module will stay within its advertised power class for all supply voltages.

For 1000kHz clock rate, refer to Figure 6-4 in [2]

IV. Optical Characteristics

- **General**

Copyright © 2026 FS.COM All Rights Reserved

Parameter	Conditions	Symbol	Min	Typ	Max	Unit	Notes
Symbol rate		R_{baud}		27.95		GBd	
Modulation format				DP-DQPSK			
Channel frequency range	100GHz grid	V_C	191.400	193.700	196.100	THz	
	50GHz grid		191.350	193.700	196.100		
Channel spacing	100GHz grid	ΔV_C		100		GHz	
	50GHz grid			50			
Frequency accuracy		δV_C	-1.8		1.8	GHz	
Side-mode suppression ratio	No modulation	SMSR	40			dB	

- **Transmitter**

Parameter	Conditions	Symbol	Min	Typ	Max	Unit	Notes
Tx output power		$P_{Tx,out}$	0			dBm	
Tx output power monitor range		$P_{Tx,mon}$	-2		4	dBm	
Tx output power monitor accuracy	Tx optical power monitor reading relative to actual Tx output power	$\delta P_{Tx,mon}$	-1.5		1.5	dB	
Tx output power during tuning or when Tx disabled		$P_{Tx,dark}$			-35	dBm	
Tx in-band optical signal to noise ratio	Under modulation, |Δf| < 60 GHz	$OSNR_{in}$	39			dB/12.5GHz	
Tx out-of-band optical signal to noise ratio	Under modulation, |Δf| > 150 GHz	$OSNR_{out}$	30			dB/12.5GHz	

- **Receiver**

Parameter	Conditions		Symbol	Min	Typ	Max	Unit	Notes
Rx total input power	Broadband		$P_{Rx,tot}$	-30		3	dBm	
Rx signal input power (amplified)	Full Rx OSNR tolerance		$P_{Rx,sig}$	-18		1	dBm	
	Extended range			-22		3		1
Rx OSNR tolerance	Back-to-back, $P_{Rx,sig}$ > -18dBm	100G DQPSK SC		16.5			dB/12.5GHz	2
		100G DQPSK RS		21.5				
CD tolerance	Default, OSNR penalty < 0.5dB					2.4	ns/nm	
	Extended, OSNR penalty < 1.0dB					6.0		3

Copyright © 2026 FS.COM All Rights Reserved

PMD tolerance	OSNR penalty < 0.5dB					10	ps	
DGD tolerance	OSNR penalty < 0.5dB					30	ps	
Tolerance to change in SOP	OSNR penalty < 0.5dB					50	krad/s	
PDL OSNR penalty	Change in principal state of polarization < 1rad/ms	1dB PDL				0.5	dB/12.5GHz	
		2dB PDL				1.0		
		4dB PDL				3.0		
Rx signal input power transient amplitude	Peak excursion from steady state, transient within Rx signal input power (amplified) range, OSNR penalty < 0.5dB			-3		3	dB	
Rx signal input power transient rise/fall time	Rise/fall time for the above peak excursion, OSNR penalty < 0.5dB			100			μs	
Colorless drop OSNR penalty	Rx total input power to signal input power ratio < 12dB					0.5	dB	4
Colorless drop adjacent channel crosstalk penalty						0.2	dB	5
Rx signal input power (unamplified)	OSNR > 35dB/12.5GHz	100G DQPSK SC		-30		1	dBm	
		100G DQPSK RS		-24		1		
Rx signal input power monitor range			$P_{Rx,mon(s)}$	-21		3	dBm	
Rx signal input power monitor accuracy			$\delta P_{Rx,mon(s)}$	-2.5		2.5	dB	
Rx total input power monitor range			$P_{Rx,mon(t)}$	-21		6	dBm	
Rx total input power monitor accuracy			$\delta P_{Rx,mon(t)}$	-2.0		2.0	dB	
Rx reflectance						-20	dB	

ⓘ **Notes:**

Rx signal input power range over which performance can be guaranteed with <1dB OSNR penalty relative to Rx OSNR tolerance limit.

Rx OSNR tolerance for Carrier Frequency Offset |CFO| < 1GHz. Up to 1dB penalty for worst case |CFO| = 3.6GHz.

Power dissipation will increase by approximately 0.2W if extended CD compensation is enabled.

Receiver is able to tolerate the specified ratio of total power of crosstalk channels and ASE to signal power with the specified OSNR penalty. Does not include contribution from adjacent channel crosstalk. No single channel power exceeding signal channel power by more than 1dB shall be included. Measured at Rx signal input power -6dBm.

Measured at the Rx OSNR limit, 50GHz channel spacing, both adjacent channels <1dB higher power than signal channel.

V. Digital Management and Diagnostics Functions

The QSFP-ZRH-100G-S QSFP28 module supports the digital management and diagnostics interface specified in the Common Management

Copyright © 2026 FS.COM All Rights Reserved

Interface Specification (CMIS) [16] with extensions specified in the OIF Coherent CMIS implementation agreement [17].

The QSFP-ZRH-100G QSFP28 module supports the diagnostics and management interface specified in the Specification for Management Interface for 4-lane Modules and Cables SFF-8636 [18], with limited control and monitoring of the coherent line interface.

VI. References

SNIA “Specification for QSFP+ 28 Gb/s 4× Pluggable Transceiver Solution (QSFP28)”, SFF-8665 Rev. 1.9 (June 29, 2015).

SNIA “Specification for QSFP+ 4× Hardware and Electrical Specification”, SFF-8679 Rev. 1.8 (October 4, 2018).

SNIA “Specification for QSFP+ 4× Module” SFF-8661 Rev. 2.5 (June 22, 2018).

SNIA “Specification for QSFP+ 4× 28 Gb/s Connector (Style A)” SFF-8662 Rev. 2.9 (June 8, 2018).

SNIA “Specification for QSFP+ 28 Gb/s Cage (Style A)” SFF-8663 Rev. 1.7 (October 19, 2017).

SNIA “Specification for QSFP+ 4× 28 Gb/s Connector (Style B)” SFF-8672 Rev. 1.2 (June 8, 2018).

SNIA “Specification for QSFP+ Cage” SFF-8683 Rev. 1.3 (October 19, 2017).

QSFP-DD MSA “QSFP-DD/QSFP-DD800/QSFP112 Hardware Specification for QSFP Double Density 8× And QSFP 4× Pluggable Transceivers” Rev. 6.3 (July 26, 2022).

IEEE Computer Society “IEEE Standard for Ethernet”, IEEE Std 802.3-2022.

ITU-T “Interfaces for the Optical Transport Network” G.709/Y.1331 Ed. 6.3 (February 2022).

ITU-T “OTU4 Long-Reach Interface” G.709.2/Y.1331.2 Ed. 1.1 (September 2020).

ITU-T “Amplified Multichannel Dense Wavelength Division Multiplexing Applications with Single Channel Optical Interfaces” G.698.2 Ed. 3.0 (November 2018).

ITU-T “Characteristics of Optical Transport Network Hierarchy Equipment Functional Blocks” G.798 Ed. 6.6 (May 2022).

CableLabs “P2P Coherent Optics Physical Layer 1.0 Specification” P2PCO-SP-PHYv1.0-I03- 200501 (May 2020).

OIF “Common Electrical I/O (CEI) - Electrical and Jitter Interoperability Agreements for 6G+ bps, 11G+ bps,25G+ bps, and 56G+ bps I/O”, OIF-CEI-4.0 (December 29, 2017).

OIF “Implementation Agreement – Common Management Interface Specification (CMIS)” OIF- CMIS-05.2 (April 2022).

OIF “Implementation Agreement for Coherent CMIS”, OIF-C-CMIS-01.3 (October 2023).

SNIA “Specification for Management Interface for 4-lane Modules and Cables”, SFF-8636 Rev 2.11 (January 3, 2023).

SNIA “Specification for SFF Module Management Reference Code Tables”, SFF-8024 Rev 4.10 (November 24, 2022).

Directive 2011/65/EU of the European Parliament and of the Council, “on the restriction of the use of certain hazardous substances in electrical and electronic equipment” as well as Commission Delegated Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU. Certain products may use one or more exemptions as allowed by the Directive.

Application Note AN-2038: “COHERENT Implementation of RoHS Compliant Transceivers”.

Test Center

I. Compatibility Testing

Each fiber optical transceiver has been tested in host device on site in FS Assured Program to ensure full compatibility with over 200 vendors.

Copyright © 2026 FS.COM All Rights Reserved

Cisco Nexus9000 C9316D-GX

Arista-DCS-7280DR3K-24-F

Juniper PTX10001-36MR

Juniper QFX5220-32cd

Dell Z9332F-ON

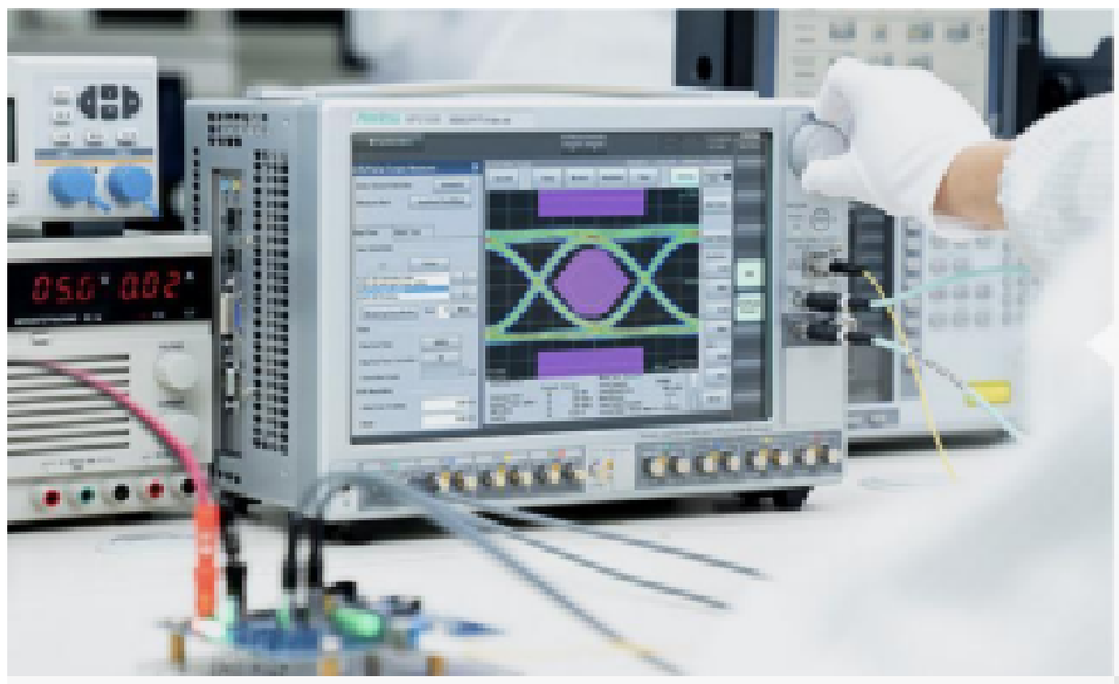
FS N9510-64D

Above is part of our test bed network equipment. For more information, please click the Test Bed PDF. It will be updated in real time as we expand our portfolio.

II. Performance Testing

Each fiber optical transceiver has been fully tested in FS Assured Program equipped with world's most advanced analytical equipment to ensure that our transceivers work perfectly on your device.

Copyright © 2026 FS.COM All Rights Reserved



1. TX/RX Signal Quality Testing

Equipped with the all-in-one tester integrated 4ch BERT & sampling oscilloscope, and variable optical attenuator to ensure the input and output signal quality.

- Eye Pattern Measurements: Jitter, Mask Margin, etc
- Average Output Power
- OMA
- Extinction Ratio
- Receiver Sensitivity
- BER Curve

2. Reliability and Stability Testing

Subject the transceivers to dramatic changes in temperature on the thermal shock chamber to ensure reliability and stability of the transceivers.

- Commercial: 0 °C to 70 °C
- Extended: -5 °C to 85 °C
- Industrial: -40 °C to 85 °C

3. Transfer Rate and Protocol Testing

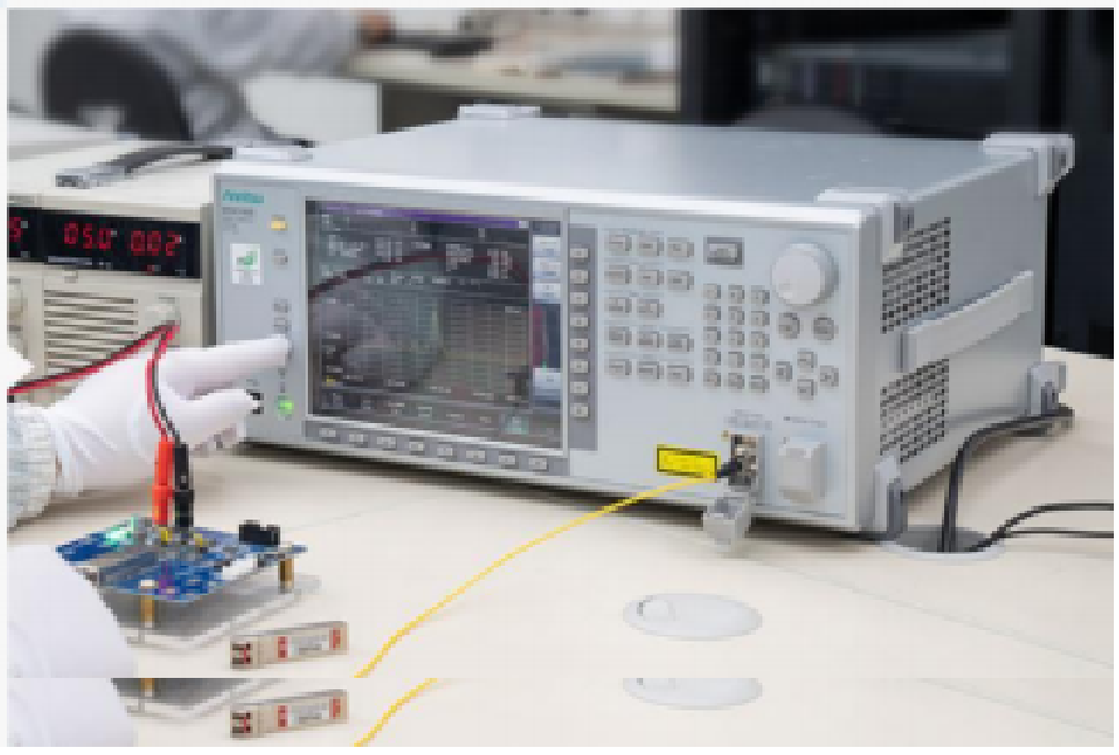
Test the actual transfer data rate and the transmission ability under different protocols with Network Master Pro.

- Ethernet
- Fibre Channel
- SDH/SONET
- CPRI

4. Optical Spectrum Evaluation

Evaluate various important parameters with the Optical Spectrum Analyzer to meet the industry standards.

- Center Wavelength, Level
- OSNR
- SMSR
- Spectrum Width



Order Information

Part Number	Description

Copyright © 2026 FS.COM All Rights Reserved

QSFP-ZRH-100G-S	100G ZR QSFP28 High-TX w/o TOF Coherent Optics Transceiver, 120km over SMF, CMIS
QSFP-ZRH-100G	100G ZR QSFP28 High-TX w/o TOF Coherent Optics Transceiver, 120km over SMF, SFF-8636
QSFP-ZRHF-100G-S	100G ZR QSFP28 High-TX with TOF Coherent Optics Transceiver, 120km over SMF, CMIS
QSFP-ZRHF-100G	100G ZR QSFP28 High-TX with TOF Coherent Optics Transceiver, 120km over SMF, SFF-8636

Copyright © 2026 FS.COM All Rights Reserved

Delaware, United States

Address：Delaware: 380 Centerpoint Blvd, New Castle, DE 19720, United States
Email：US@fs.com
Tel：+1 (888) 468-9910

California, United States

Address：California: 15241 Don Julian Rd, City of Industry, CA 91745, United States
Email：US@fs.com
Tel：+1 (888) 468-9910

Germany

Address：Röntgenstraße 18, 85757 Karlsfeld, Germany
Email：DE@fs.com
Tel：+49 (0) 8131 377 3008

United Kingdom

Address：Unit 8, Urban Express Park, Union Way, Aston, Birmingham B6 7FH, United Kingdom
Email：UK@fs.com
Tel：+44 (0) 121 726 4775

Australia

Address：57-59 Edison Rd, Dandenong South, VIC 3175, Australia
Email：AU@fs.com
Tel：+61 3 5909 9990

Singapore

Address：7002 ANG MO KIO AVENUE 5 #05-02 Singapore 569914
Email：SG@fs.com
Tel：+65 31381992

Japan

Address：JS Progress Building 5F, 4-1-23, Heiwajima, Ota Ku, Tokyo, 143-0006, Japan
Email：JP@fs.com
Tel：+81-3-6897-9438

FS has several offices around the world. Addresses, phone numbers are listed on the FS Website at https://www.fs.com/contact_us.html. FS and FS logo are trademarks or registered trademarks of FS in the U.S. and other countries.

Copyright © 2026 FS.COM All Rights Reserved